

Infocare CRM — Technical Specification

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1. Overview

Infocare will build its own CRM to capture leads from direct marketing, email, social media, WhatsApp and the website, and carry each one to a won deal and a handover to accounting. Target: a working first release (leads, pipeline, field app) within 3 weeks of development start, and quotations within 6 weeks.

In scope: leads and contacts, pipeline, activities and tasks, product list and price list, quotations/BOQ with PDF, AI assistance, dashboards, and a sync to the separate accounting system.

Out of scope for now: accounting, inventory, project execution and payroll. These stay in existing tools and connect later.

Users:

Group	Location	Main use
Management	Group	Dashboards, approvals, reports
Sales and marketing	Dubai, RAK	Leads, visits, pipeline, follow-ups
Back office	Kerala	Quotations, BOQs, drawings, payment follow-ups
Admin	Group	Users, product list, settings

Design principles:

1. The database is the single source of truth. Prices, totals and VAT are always calculated by the system.
2. AI drafts and suggests; a person approves anything a customer sees.
3. Every lead keeps its source and campaign from the moment it is created.
4. Configure and automate before writing custom code; keep the codebase small enough for a small team to maintain.

2. Tech stack and environments

The app is built in Flutter on Supabase (PostgreSQL). One Flutter codebase produces the Android and iOS field app plus a Windows desktop and web version for the office; the database, n8n workflows and AI features are unchanged.

Layer	Choice	Notes
Database	PostgreSQL via Supabase	Auth, storage, row-level security; self-hostable
App, all devices	Flutter 3 (Dart), Material 3	Android + iOS for field sales; Windows desktop or web for the office
State and navigation	Riverpod + go_router	Widely used, well documented
Data access	supabase_flutter SDK	Login, queries, file upload, live updates
Offline on the phone	Local SQLite (drift), synced on reconnect	Visits and photos saved without signal
Tables and charts	data_table_2 (or a data-grid package), fl_chart	Large lists, dashboard charts
Server logic	Supabase Edge Functions (TypeScript) + Postgres functions	Pricing, numbering, PDF, Claude calls
Automation	n8n, self-hosted (Docker)	Unchanged
AI	Claude API via Edge Functions or n8n	Never called from the app
Push notifications	Firebase Cloud Messaging	New lead, overdue follow-up
Hosting	Docker Compose on a VPS for Supabase + n8n	Apps via Play Store, App Store, web hosting
Code and builds	Git + GitHub Actions or Codemagic	Automated Android, iOS, Windows, web builds

What changes compared with the Next.js plan:

- **Server logic moves to Edge Functions.** Flutter has no server of its own, so pricing, numbering, PDF and AI calls run in Supabase.
- **Security rests on row-level security.** Every device talks to the database directly, so each role's rules must be tested.
- **The phone app becomes native.** Camera, GPS, offline storage and push alerts work better than in a browser app.
- **Phone releases go through the app stores.** Review usually takes a few days; office desktop and web updates are immediate.
- **Flutter web is slower on first load** and weaker for very large tables; the Windows desktop build avoids both for office staff.
- **Unchanged:** data model, roles, n8n workflows, AI features and screen flow.

Environments: local (Docker), staging (plus TestFlight and Play internal testing), production.

Team note: Flutter covers the app only. Pair the Flutter developer with one person for SQL, Edge Functions and n8n.

3. Data model

Fourteen core tables cover release 1 and 2. Every table has id (UUID), branch_id, created_at, created_by, updated_at; money is stored as numeric(12,2) in AED.

Table	Key fields	Links to
branches	name (Dubai, RAK, Kerala), currency, VAT no.	—
users	name, email, role, branch_id, active	branches
companies	name, trade licence no., industry, address, emirate	—
contacts	name, mobile, email, WhatsApp no., position	companies
campaigns	name, channel, start/end date, budget	—
leads	title, requirement text, product_interest[], source , source_detail, campaign_id, status, owner_id, ai_summary, ai_category, duplicate_of	contacts, companies, campaigns, users
deals	name, stage, value, probability, expected_close, lost_reason, won_at	leads, companies, users
activities	type (call, visit, email, WhatsApp, note), summary, occurred_at, gps, attachments	leads or deals, users
tasks	title, due_at, status, assignee_id, reminder_sent	leads or deals, users
products	SKU, name, brand, model, category, unit, cost, list_price, active, datasheet_url	—
product_bundles	name, lines (product_id, qty)	products
quotations	number, version, status, valid_until, subtotal, vat, total, terms, ai_drafted, approved_by, pdf_url	deals, contacts
quotation_lines	product_id, description, qty, unit_price, discount, line_total, sort	quotations, products
audit_log	table, record_id, action, old/new values, user_id	all

Fixed values (enums):

- **source:** direct_visit, event, walk_in, referral, email, facebook, instagram, linkedin, whatsapp, website, phone.
- **lead status:** new, contacted, qualified, disqualified, converted.
- **deal stage:** site_survey, quotation_sent, negotiation, won, lost.

- **quotation status:** draft, pending_approval, approved, sent, accepted, rejected, expired.

Rules enforced in the database:

- Line totals, subtotal, 5% VAT and total are computed by a Postgres function; clients cannot write them.
- Quotation numbers follow Q-YYYY-NNNN from a sequence; a revision keeps the number and increments version.
- A lead is flagged as a duplicate when its mobile or email matches an existing contact.
- The product list is loaded first from the product Excel list prepared by Anand and Sujith.

4. Roles and permissions

Five roles, enforced with Postgres row-level security on branch_id and owner_id, so the rules hold even if the app has a bug.

Role	Sees	Can change	Approves
Admin	Everything	Users, products, prices, settings	—
Management	All branches	Anything except settings	Quotations, discounts, lost reasons
Branch sales	Own branch	Own leads, deals, activities, draft quotations	—
Back office (Kerala)	All branches' deals and quotations	Quotations, BOQs, drawings, tasks	—
Viewer	Dashboards and reports	Nothing	—

- A discount above a set limit (to be decided) sends the quotation to management approval.
- Price list and product cost are editable by Admin only; cost is hidden from Branch sales.
- Every change to a deal, quotation or price is written to audit_log.

5. Lead intake by channel

All channels land in one leads table through one n8n intake workflow, so every lead gets the same checks, AI summary and owner assignment.

Channel	How it arrives	Source value	Notes
Direct marketing	Flutter phone app form, saved straight to the database	direct_visit, event, walk_in	GPS and photos attached; works offline and syncs later
Email	n8n reads the shared sales inbox (Gmail or Microsoft 365)	email	Only new threads create leads; replies attach as activities
Facebook / Instagram ads	Meta Lead Ads webhook to n8n	facebook, instagram	Campaign ID mapped to campaigns
Facebook / Instagram messages	Meta Messenger / Instagram messaging webhook	facebook, instagram	Needs a Meta business app and page access
LinkedIn	Lead Gen Forms via export or a sync tool	linkedin	LinkedIn API access is restricted; start with CSV import
WhatsApp	WhatsApp Business Platform (Cloud API) webhook	whatsapp	Needs a verified business number; message templates for outbound
Website	Form posts to an n8n webhook	website	Add hidden fields for UTM campaign tags
Phone / referral	Entered manually in the web app	phone, referral	Referrer stored in source_detail

Intake steps (every channel):

1. Normalise the phone number to +971 format and lower-case the email.
2. Check for an existing contact; if found, attach an activity instead of creating a new lead.
3. Send the enquiry text to Claude for a summary, product category and urgency.
4. Assign an owner by branch (emirate or number prefix) and product category, using a round-robin within the branch.
5. Create a task: respond within 2 working hours (target to confirm).
6. Notify the owner by email or WhatsApp.

6. n8n workflows

Ten workflows run the automation; each is exported to the Git repository as JSON and has a named owner.

#	Workflow	Trigger	What it does	Release
W1	Lead intake	Webhooks, email poll (every 5 min)	Section 5 steps: clean, de-duplicate, AI summary, assign, notify	1
W2	Response SLA	Every 15 min	New lead untouched past target → remind owner, then escalate to manager	1
W3	Follow-up reminders	Daily 8:00	Tasks due today and overdue → one message per person	1
W4	Daily briefing	Weekdays 8:30	Pipeline changes since yesterday → Claude writes a short briefing for the 8:45 call	1
W5	Quote chaser	Daily 10:00	Quotation sent, no reply for 5 days → task + AI draft follow-up for approval	2
W6	Quote PDF	Quotation approved	Render PDF, store it, email the salesperson the link	2
W7	Pipeline review report	Mon and Thu 8:00	New, stalled, won and lost deals by branch and owner	2
W8	Accounting sync	Deal won; hourly for payments	Create customer + invoice draft in accounting; pull payment status back	4
W9	AMC renewals	Daily	Contracts ending in 60/30/7 days → renewal deal + task	4
W10	Health check	Every 30 min	Failed workflow runs, API errors → alert admin	1

Conventions:

- n8n writes to the database only through the API with a service account, never with raw SQL.
- Every workflow logs its run ID on the records it touches.
- Secrets (API keys, tokens) live in n8n credentials, never in workflow JSON.

7. AI features and guardrails

Claude is used for six jobs, each with a fixed prompt kept in the repository and a structured (JSON) output that the system validates before saving.

Feature	Input	Output	Human step
Lead summary and category	Enquiry text	Summary, product category, urgency, missing info	None; editable on the lead
Reply and follow-up drafts	Lead, last activities, quotation status	Email or WhatsApp draft	Approve, edit or discard
BOQ draft	Survey notes, photo captions, product list (IDs only)	Product IDs + quantities + reasoning	Back office reviews every line
Daily briefing	Pipeline changes, overdue tasks	5–8 line summary	None (internal)
Voice note to text	Audio from the field app	Transcript + extracted fields	Salesperson confirms
Product Q&A	Question + data sheets	Answer with the source document	Internal use only

Guardrails:

1. The AI returns product IDs and quantities only; the system looks up prices and calculates totals.
2. Nothing is sent to a customer without a person clicking Approve.
3. Only the fields a task needs are sent; no cost prices, no other customers' data.
4. Every AI call is logged (feature, record, tokens, approved or edited).
5. The API key sits in Edge Functions or n8n, never inside the app.
6. If the AI call fails, the workflow continues without it; AI never blocks a lead from being saved.

Monthly API cost should be modest at this volume; set a spending limit in the Anthropic console and review the usage log monthly.

8. Screens and functional requirements

Each screen matches an artboard in the design canvas; the list below is what "done" means for each.

Dashboard

- KPI tiles: new leads this week, open pipeline value, quotes awaiting reply, follow-ups due.
- Leads by source for a chosen period; filter by branch.
- Morning briefing (from W4) and today's follow-ups.

Lead inbox

- One list for all channels; filter by source, branch, owner, status, date.
- Source tag, AI requirement summary and age on every row; overdue rows highlighted.
- Bulk assign, CSV import and export.

Lead detail

- Contact card, source and campaign, stage tracker, activity timeline.
- Log call, visit, note; add task; convert lead to deal.
- AI panel: next step and draft reply with Approve / Rewrite.

Pipeline

- Board by stage with value per column; drag to move stage.
- Moving to Lost requires a reason; moving to Won triggers W8.
- Cards show days in stage; stalled deals highlighted.

Quotation / BOQ builder

- Add lines from the product list or bundles; edit qty and discount.
- AI draft from survey notes, clearly labelled.
- Totals and VAT computed by the system; revisions keep history.
- Send for approval → approved → PDF → send.

Field app (phone)

- Log a visit in under a minute: company, contact, interest, notes, photos, voice note.
- Offline save and sync; GPS captured.
- My tasks today.

Admin

- Users and roles, branches, product list import, campaigns, lost reasons, SLA targets.

9. Integrations

Every outside system connects through n8n; the CRM itself never calls them directly.

System	Direction	Data	Owner of the data
Accounting (separate)	CRM → accounting on Won; accounting → CRM hourly	Customer, invoice draft from the approved quotation; payment status and balance back	Customers and products: CRM. Invoices and payments: accounting
Email (Gmail or Microsoft 365)	Both	Inbound enquiries; outbound approved messages from a shared address	CRM stores a copy as an activity
WhatsApp Business Platform	Both	Inbound messages; outbound approved templates	CRM stores a copy as an activity
Meta (Facebook, Instagram)	Inbound	Lead Ads forms and page messages	CRM
Claude API	Outbound calls	Section 7 jobs	—

The accounting connector depends on which accounting software is chosen (open decision 1). Build W8 last, against its API or an import file.

10. Security, backup and performance

- **Access:** email + password with two-factor authentication for Admin and Management; sessions expire after 12 hours.
- **Data isolation:** row-level security on every table (section 4); tested with automated tests per role.
- **Backups:** nightly full database backup, 30 days kept, one copy off the server; restore tested monthly on staging.
- **Files:** photos, drawings and PDFs in private storage, served by short-lived signed links.
- **Secrets:** in environment variables and n8n credentials only; rotated when a staff member leaves.
- **Updates:** Supabase, n8n and the OS patched monthly on staging first.
- **Monitoring:** uptime check, error alerts (W10), disk and backup alerts.
- **Performance targets:** lists load in under 2 seconds with 50,000 leads; field app usable on 4G.
- **Privacy:** customers' personal data used only for sales and service; an export and delete process on request.

11. Delivery plan and task list

Four releases over about 10 weeks from development start; each ends with a demo to the team and goes live only after it passes on staging.

A usable CRM in 3 weeks, full scope in about 10



Each release ends with a team demo and a staging sign-off before it goes live.

Release 1 is the one that matters most: once leads and the pipeline are live, the team stops using spreadsheets.

Week 0 — Setup

- Create Git repository, staging and production servers
- Install Supabase and n8n with Docker; set up backups; open Apple and Google Play developer accounts and a Firebase project for push
- Create database schema and migrations (section 3)
- Import product list from the product Excel file
- Import existing projects list as companies and won deals

Release 1 (weeks 1–3) — Leads and pipeline

- Login, roles and row-level security (section 4)
- Lead inbox, lead detail, activities and tasks
- Pipeline board
- Phone app (Android first, then iOS): my day, log a visit, photos, offline sync, push alerts
- W1 intake for website form and email; W2, W3, W4, W10
- AI lead summary and category

Release 2 (weeks 4–6) — Quotations

- Quotation builder, bundles, revisions, approval step
- Quotation PDF template and W6
- AI BOQ draft and follow-up drafts; W5, W7

Release 3 (weeks 7–8) — More channels

- WhatsApp Business Platform intake and approved replies
- Meta Lead Ads and page messages; LinkedIn CSV import
- Voice note to text in the field app
- Dashboard and source/campaign reports

Release 4 (weeks 9–10) — Accounting and renewals

- W8 accounting sync and payment status
- W9 AMC renewals
- Staff training and user guide